```
Calibration results
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Normalized Residuals
----------------------------
Reprojection error (cam0):     mean 0.320219008467, median 0.296407520353, std: 0.174568103953
Reprojection error (cam1):     mean 0.302291398114, median 0.276414108088, std: 0.170783244372
Gyroscope error (imu0):        mean 0.660610852276, median 0.575949037574, std: 0.398085864999
Accelerometer error (imu0):    mean 0.412044952352, median 0.352147255654, std: 0.283891206682

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 0.320219008467, median 0.296407520353, std: 0.174568103953
Reprojection error (cam1) [px]:     mean 0.302291398114, median 0.276414108088, std: 0.170783244372
Gyroscope error (imu0) [rad/s]:     mean 0.00952746520403, median 0.00830645514811, std: 0.0057412759
Accelerometer error (imu0) [m/s^2]: mean 0.0594259681401, median 0.050787399471, std: 0.04094337209

Transformation (cam0):
-----------------------
T_ci:  (imu0 to cam0):
[[ 0.99998697  0.00474593 -0.00188255 -0.03644422]
 [ 0.00474361 -0.99998799 -0.00123547 -0.00379884]
 [-0.00188839  0.00122653 -0.99999746 -0.02581118]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0):
[[ 0.99998697  0.00474361 -0.00188839  0.03641302]
 [ 0.00474593 -0.99998799  0.00122653 -0.00359417]
 [-0.00188255 -0.00123547 -0.99999746 -0.02588442]
 [ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.00899697182225


Transformation (cam1):
-----------------------
T_ci:  (imu0 to cam1):
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```
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[ 0.99998783 -0.0000127  -0.00493403 -0.00335636]
 [-0.00000803 -0.99999955  0.00094715 -0.00351876]
 [-0.00493404 -0.00094709 -0.99998738 -0.02588016]
 [ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.00901920440792

Baselines:
----------
Baseline (cam0 to cam1):
[[ 0.99998404  0.00475767  0.00304565  0.03976892]
 [-0.00475684  0.99998865 -0.00027941  0.00007491]
 [-0.00304694  0.00026492  0.99999532 -0.00019203]
 [ 0.          0.          0.          1.        ]]
baseline norm:  0.0397694570886 [m]


Gravity vector in target coords: [m/s^2]
[-0.02531681 -0.04815697 -9.80639908]


Calibration configuration
=========================

cam0
-----
  Camera model: pinhole
  Focal length: [122.75919056803501, 122.46026535001319]
  Principal point: [328.13414733341733, 239.01585045965925]
  Distortion model: equidistant
  Distortion coefficients: [0.45331106231325524, 0.2678213747893726, -0.10766023881714477, 0.0149351
  Type: aprilgrid
  Tags:
```

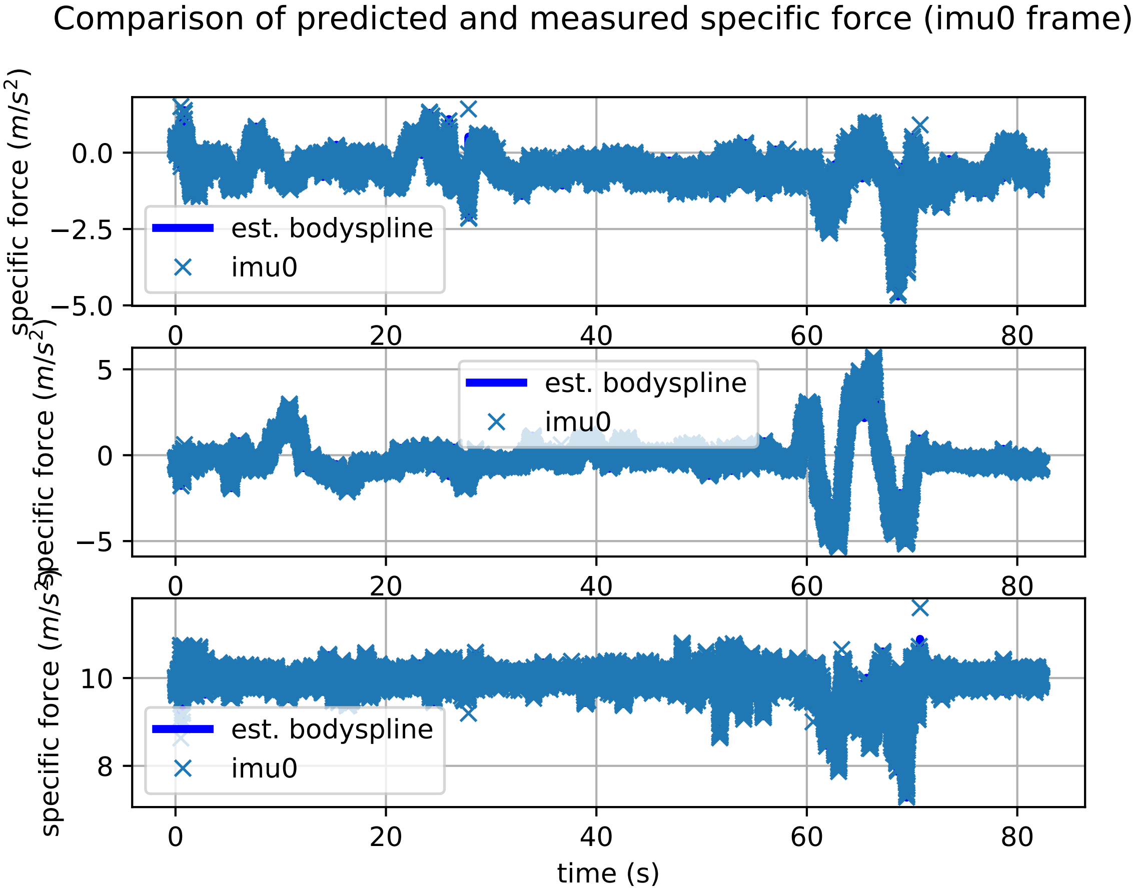
```
    Spacing 0.010372 [m]


cam1
-----
  Camera model: pinhole
  Focal length: [121.8861244558415, 121.71189157089208]
  Principal point: [319.1312214501519, 242.66971178285496]
  Distortion model: equidistant
  Distortion coefficients: [0.4893430117644999, 0.17093148878742823, -0.008274679023642487, -0.02041
  Type: aprilgrid
  Tags:
    Rows: 7
    Cols: 5
    Size: 0.02593 [m]
    Spacing 0.010372 [m]

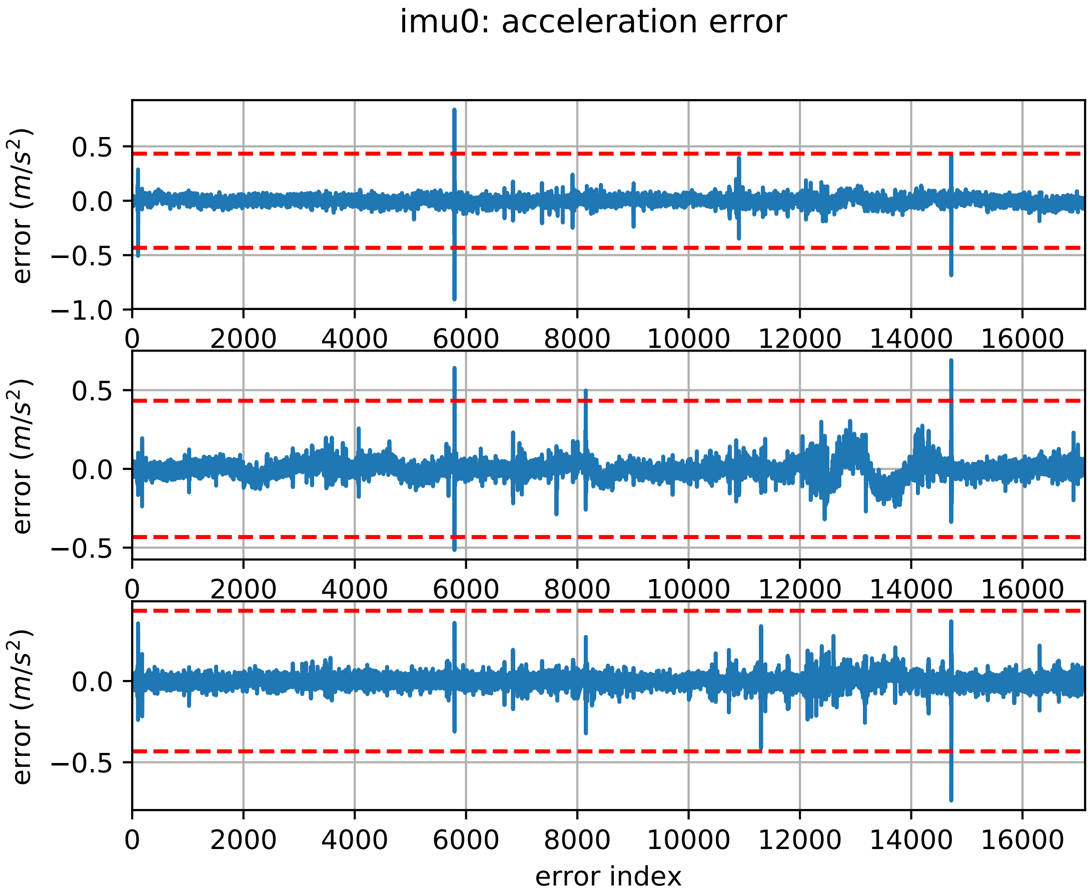


IMU configuration
=================

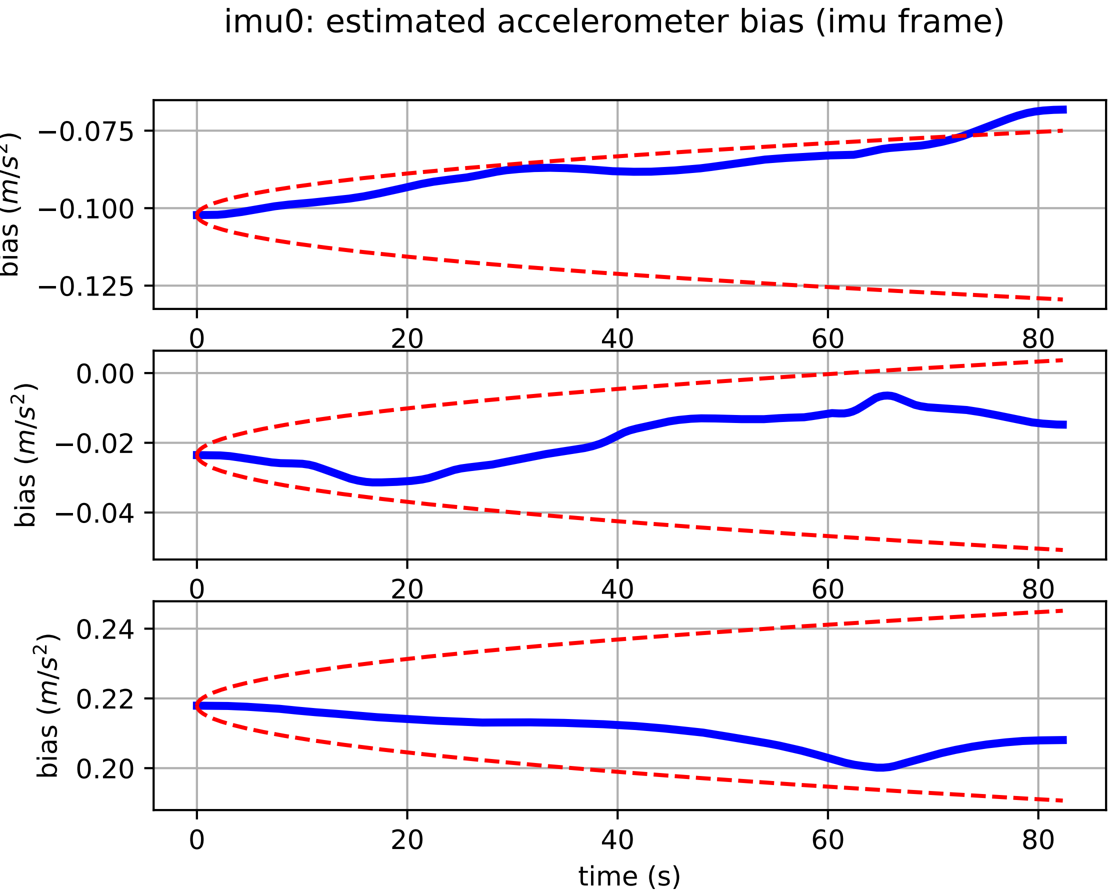
IMU0:
----------------------------
  Model: calibrated
  Update rate: 208.0
  Accelerometer:
    Noise density: 0.01
    Noise density (discrete): 0.144222051019
    Random walk: 0.001
  Gyroscope:
    Noise density: 0.001
    Noise density (discrete): 0.0144222051019
    Random walk: 0.0001
  T_i_b
    [[ 1.  0.  0.  0.]
     [ 0.  1.  0.  0.]
```

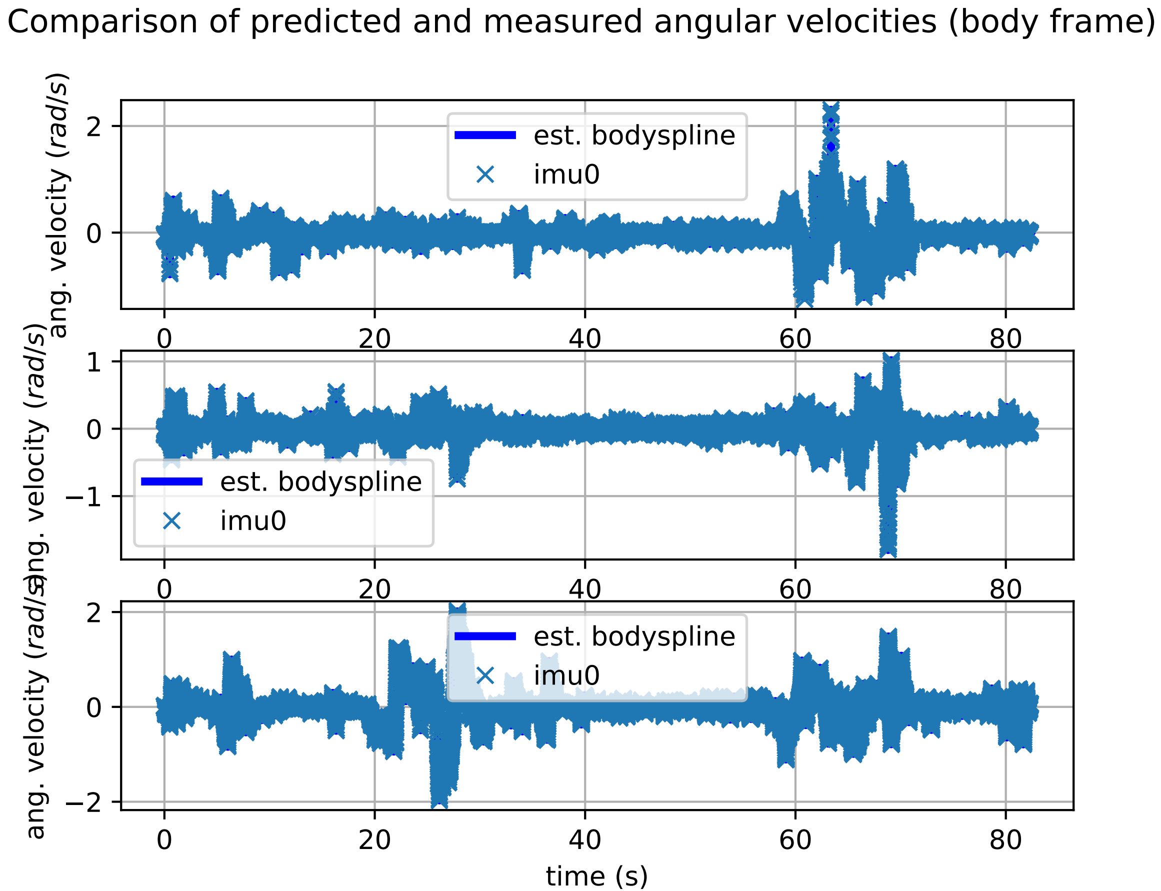


Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
0.0
−2.5
−5.0
0
20
40
60
80
est. bodyspline
imu0
5
0
−5
10
8
time (s)

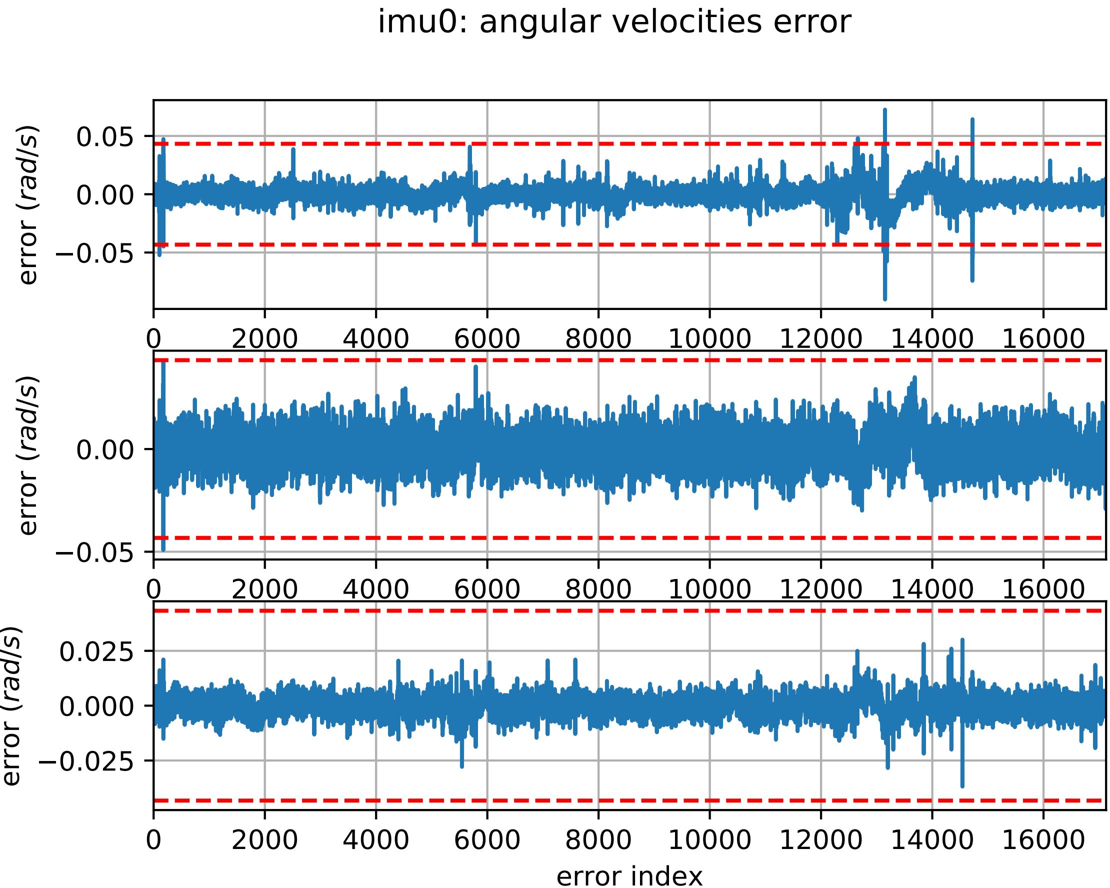


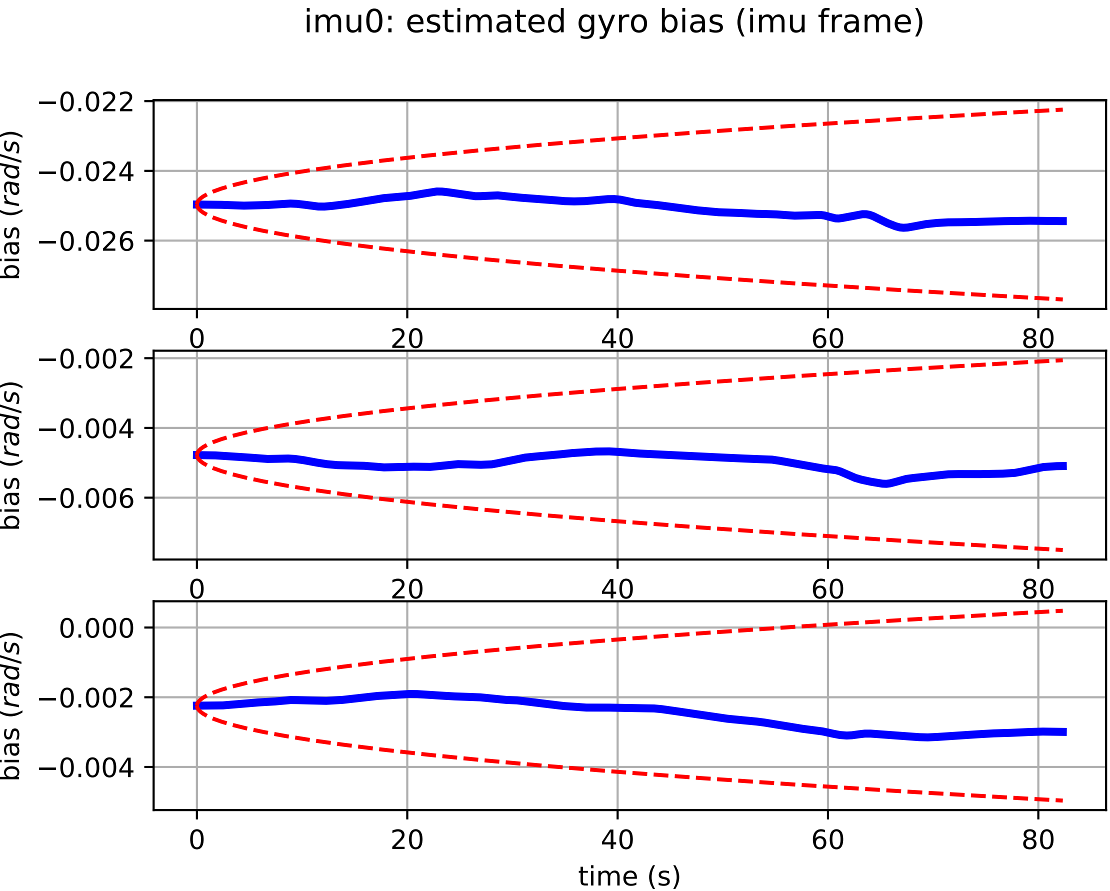
imu0: acceleration error
error (m/s^2)
error (m/s^2)
error (m/s^2)
0.5
0.0
−0.5
−1.0
−0.5
0
2000
4000
6000
8000
10000
12000
14000
16000
error index

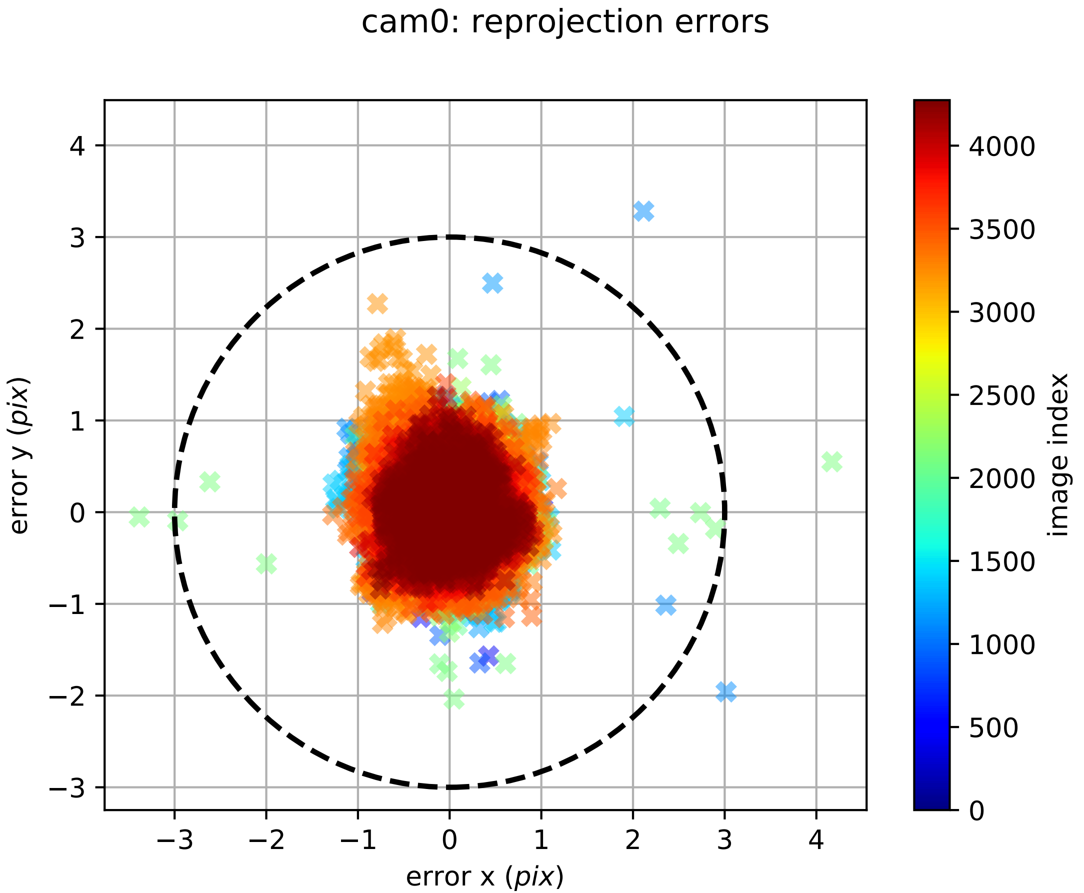

imu0: estimated accelerometer bias (imu frame)
−0.075
−0.100
−0.125
bias (m/s^2)
0
20
40
60
80
0.00
−0.02
−0.04
bias (m/s^2)
0
20
40
60
80
0.24
0.22
0.20
bias (m/s^2)
0
20
40
60
80
time (s)

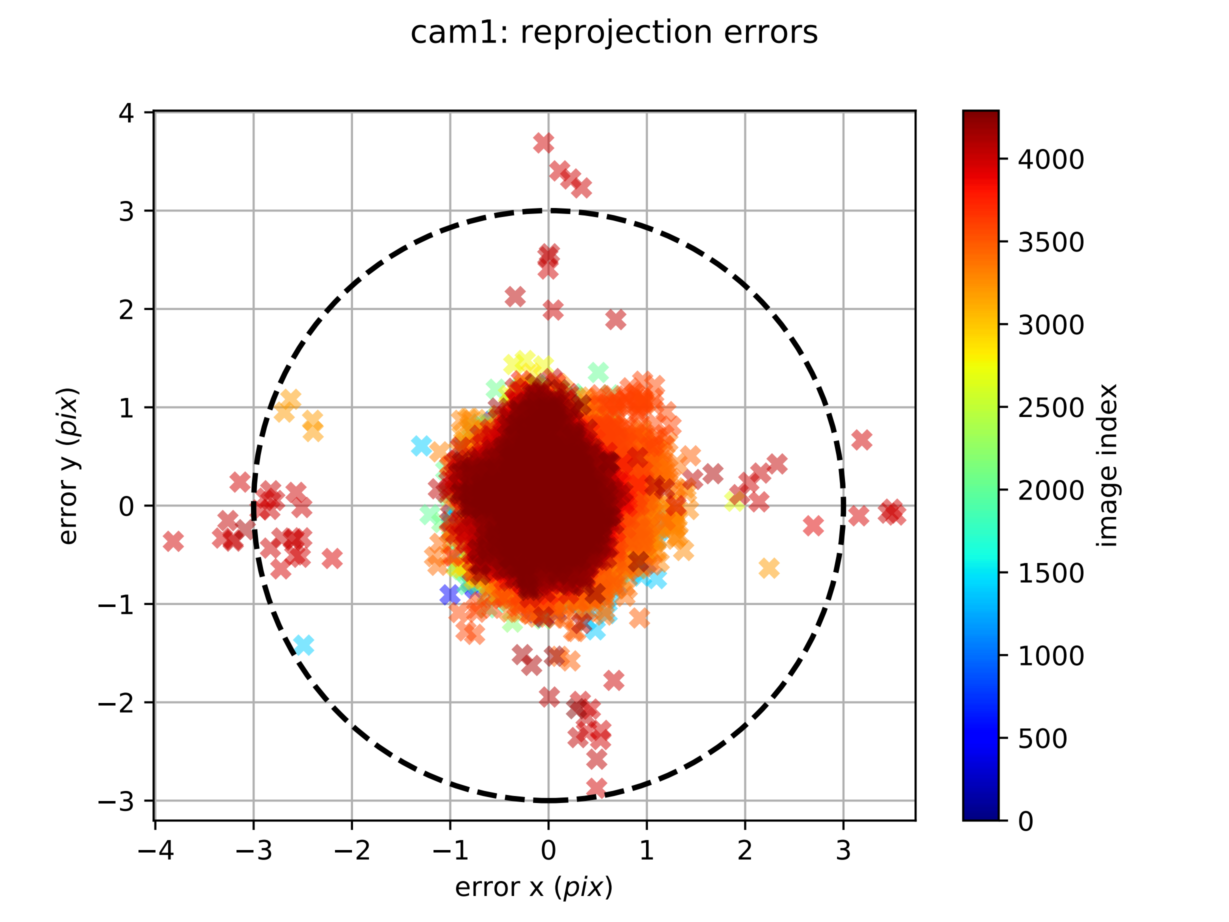


Comparison of predicted and measured angular velocities (body frame)
est. bodyspline
imu0
ang. velocity (rad/s)
2
0
1
−1
−2
0
20
40
60
80
time (s)


imu0: angular velocities error
error (rad/s)
0.05
0.00
−0.05
error (rad/s)
0.00
−0.05
error (rad/s)
0.025
0.000
−0.025
0
2000
4000
6000
8000
10000
12000
14000
16000
error index


imu0: estimated gyro bias (imu frame)
−0.022
−0.024
−0.026
bias (rad/s)
0
20
40
60
80
−0.002
−0.004
−0.006
bias (rad/s)
0
20
40
60
80
0.000
−0.002
−0.004
bias (rad/s)
0
20
40
60
80
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
4
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3
4
0
500
1000
1500
2000
2500
3000
3500
4000


cam1: reprojection errors
error y (pix)
error x (pix)
image index
4
3
2
1
0
−1
−2
−3
−4
4000
3500
3000
2500
2000
1500
1000
500